ARAB YOUTH SPACE HACKATHON 2026 · 813 CHALLENGE · PROOF OF CONCEPT

BenthosAI
Seabed depth and water clarity from space

Theme: Water Quality & Inland/Coastal Water Intelligence

Abu Dhabi Maritime Academy · AD Ports Group

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Ships are most at risk where the water is least surveyed

1

Hardest water to survey

Shallow coastal water is costly and unsafe for survey boats, and its seabed shifts after storms and dredging.

2

Out-of-date charts

Many coastal charts are decades old; channels silt up between surveys.

3

Turbidity goes unmeasured

Turbid water hides habitats and hazards, and nobody maps it routinely.

Who is affected

- **Port authorities and harbour masters:** safe depth in channels and berths
- **Hydrographic offices:** which chart areas to re-survey first
- **Dredging contractors:** siltation between surveys
- **Environment agencies:** water clarity over seagrass and coral

A fast, honest first look before sending a boat

1

End user

Port hydrographer or coastal manager planning surveys and monitoring change

2

Decision

Where to survey first · has the channel silted · is the water clear · where can the map be trusted

3

Today

Vessel surveys: accurate, but days to weeks and costly per area; charts updated rarely

What BenthosAI delivers

- **A dated depth map with a 95 % uncertainty for every pixel** and an IHO S-44 grade, showing where a vessel survey is still needed
- **A turbidity (water-clarity) map** from the same images: water-quality monitoring at no extra cost
- **Free data, minutes on a laptop**, repeatable every ~5 days

Only free, public satellite data in the proof of concept

Dataset	What it gives	Used here
Sentinel-2 L2A (ESA Copernicus)	Multispectral images, 10 m, every ~5 days	3 dates, Aug–Sep 2022, tile T39QXG (Microsoft Planetary Computer)
ICESat-2 ATL24 (NASA)	Green-laser seabed depths along tracks	7 passes 2019–2023 on 3 tracks: calibration and independent test
EOT20 tide model (DGFI-TUM)	Global ocean tides	Laser depths reduced to LAT (chart datum)
Hyperspectral (EnMAP / PRISMA)	Hundreds of narrow bands	Not used in the PoC; next step for seabed type, chlorophyll and sediment

Confidential training data stay private. Our production model is also trained on ≈240,000 AD Ports Group multibeam soundings. These soundings and the trained weights are not in the repository; everything a reviewer runs uses public data.

From image to depth, uncertainty and turbidity

1

Clean

Reflectance, clouds (SCL), water mask (NDWI + SWIR), sun-glint removal

2

Water quality

NDTI turbidity, red-edge sediment signal, “seabed visible?” test

3

Features

Stumpf log-ratio and Lyzenga log-bands

4

Calibrate

On ICESat-2 depths, one fit per image date

5

Fuse + U95

3 dates weighted; conformal 95 % interval per depth band

6

Test

Unseen laser tracks; RMSE, bias, IHO S-44

Leak-free testing

- ICESat-2 repeats the same ground track, so we split by track and swap roles (2 folds): 1218 ↔ 0006 + 0448
- A 100 m buffer removes neighbouring pixels, so no information leaks

Honest outputs

- Every pixel carries depth, ±U95 and a flag: valid · turbid · optically deep · land/cloud
- Unreadable water is flagged, not guessed

Physics-based features and gradient boosting

Inputs: 7 features per pixel

- **pSDB**: Stumpf ln(1000·R_blue) / ln(1000·R_green)
- **X_B01…X_B04**: Lyzenga ln(R_i − R∞,i), deep-water signal removed
- **B05 turbidity**: red-edge sediment signal
- **NDTI**: (red − green)/(red + green)
- *No coordinates, so the model reads water colour instead of memorising a place*

Learner: gradient-boosted trees

- scikit-learn **HistGradientBoostingRegressor**
- 300 trees · learning rate 0.05 · ≤ 15 leaves · ≥ 10 pixels per leaf · L2 1.0 · seed 42
- **Depth-balanced weights** (0–2, 2–5, > 5 m; capped ×5), so deeper water is not swamped by shallows
- Target: ICESat-2 ATL24 depth below LAT (≥ 3 photons, confidence ≥ 0.6, spread ≤ 0.5 m)

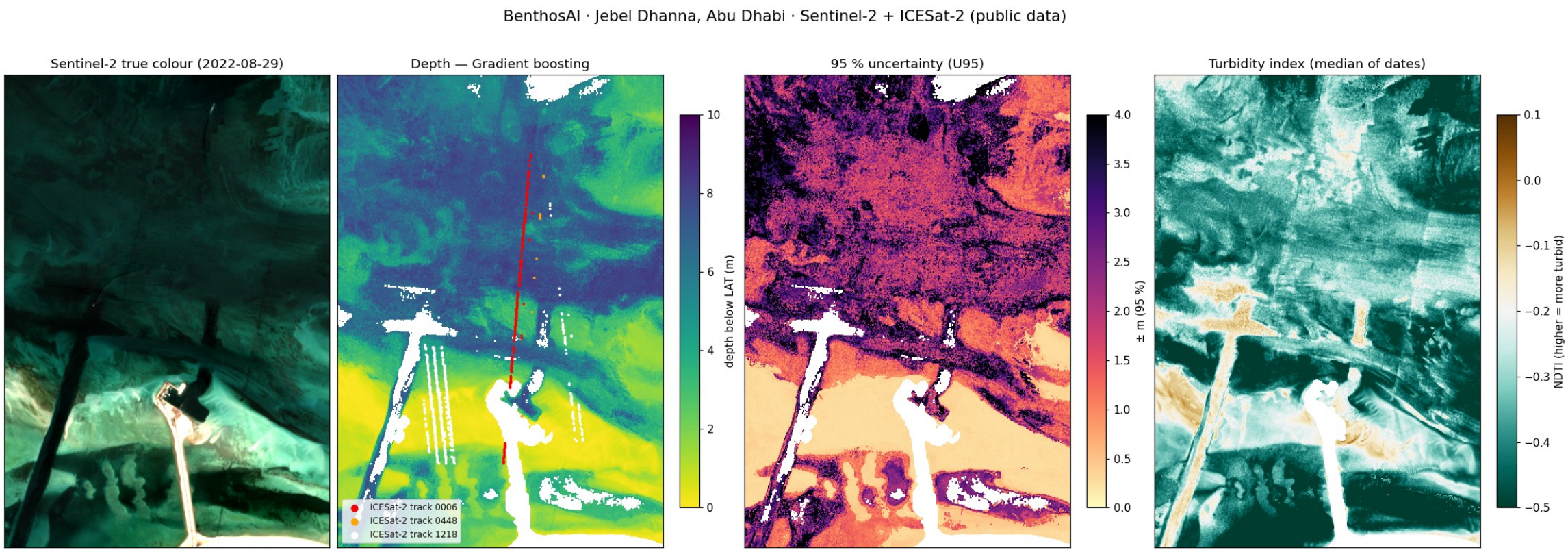
Calibration, fusion, uncertainty

- **One fit per image date** absorbs that day's tide, haze and water colour
- 3 dates fused, weighted by out-of-sample error (leave one beam pair out)
- **U95 = q_band·(σ + γ)**: σ = spread between dates; split-conformal per depth band
- Baselines: Stumpf and Lyzenga (robust Huber regression)

Production pipeline

Pipeline v1.4.3 (weights confidential, available to the committee on request): LightGBM on 13 physics-based features with monotone (physical) constraints, trained on ≈240,000 AD Ports multibeam soundings at 7 sites and tested leave-one-site-out; a U-Net ensemble is kept as a challenger.

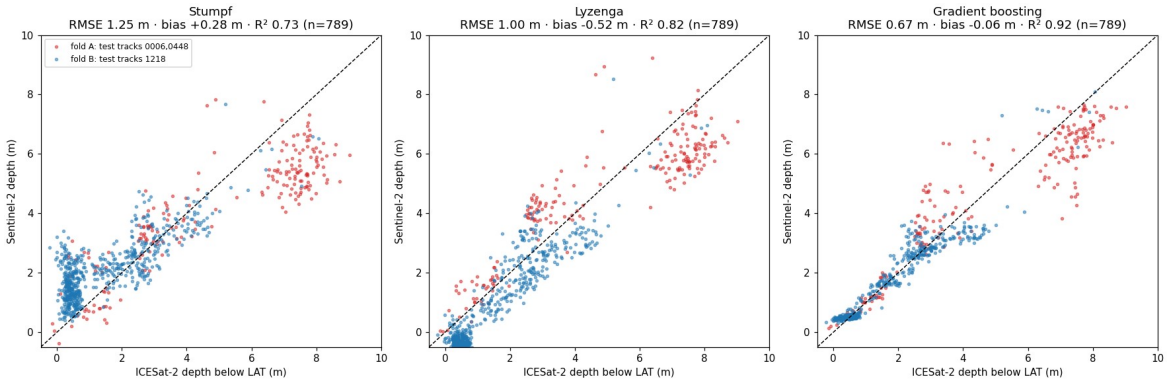
Jebel Dhanna, Abu Dhabi: depth, uncertainty and turbidity



True colour · depth below LAT · 95 % uncertainty · NDTI turbidity. ICESat-2 tracks: 1218 white, 0006 red, 0448 orange. Output of notebooks/02_main_analysis.ipynb (final model trained on all tracks).

Tested on laser tracks the models never saw

0.67 m	0.92	86 %	97 %
RMSE, UNSEEN TRACKS	R^2	IHO S-44 ORDER 2	U95 COVERAGE



Predicted vs ICESat-2 depth, cross-track (789 pixels)

Limitations

Optical reach ≈ 10–12 m in clear water; less when turbid (flagged)

Laser passes 2019–2023 vs images 2022; the dredged site has changed

Reference is satellite laser, not a certified survey; not for navigation

Model	RMSE	MAE	Bias	R^2	Ord 2	U95 cov.
Stumpf	1.25	1.00	+0.28	0.73	59 %	99 %
Lyzenga	1.00	0.86	-0.52	0.82	67 %	90 %
Gradient boosting	**0.67**	**0.42**	**-0.06**	**0.92**	**86 %**	**97 %**

Gradient boosting	n	RMSE m	Bias m	Ord 2
Depth 0–2 m	431	0.20	+0.09	100 %
Depth 2–5 m	243	0.77	+0.10	81 %
Depth 5–12 m	115	1.29	-0.97	46 %
Fold A (test 0006+0448)	207	1.11	-0.28	63 %
Fold B (test 1218)	582	0.41	+0.02	95 %

Production model: internal validation

- Confidential: ≈240,000 multibeam soundings, 7 Abu Dhabi sites, whole sites held out
- Khalifa Port 2021: **RMSE 0.26 m** · S-44 Order 2 99.8 %
- All sites, < 12 m (38,345 pts): RMSE 0.85 m · bias −0.02 m

Why it matters

~5 days	$0 data	±U95
REVISIT BETWEEN VESSEL SURVEYS	FREE SATELLITE AND TIDE DATA	TRUST LEVEL ON EVERY PIXEL

What this changes

- **Safer navigation:** spot shoaling in channels and approaches early
- **Lower survey cost:** send vessels only to uncertain or changing areas
- **Water quality:** turbidity maps for dredging plumes, habitats and coastal works from the same images
- **Regional scale:** any clear Arab-region coast (Gulf, Red Sea) with ICESat-2 coverage; runs on a laptop in minutes

9 NEXT STEPS

From proof of concept to an operational service

Technical

- **Hyperspectral** (EnMAP / PRISMA) for seabed type, chlorophyll and sediment
- Time series: depth and turbidity change maps for siltation monitoring
- More sites across the Gulf and Red Sea; automatic tide reduction everywhere
- Simple web app: draw an area → maps and report (desktop prototype exists)

Partnerships

- Pilot with port and hydrographic partners in the UAE
- Grade against IHO S-44 with independent surveys
- Compute and data support for regional scale-up

Reproducible today: notebook, sample data, results and README run end to end in under a minute on a laptop

Thank you

Dr Muhammad Saeed	Team lead: SDB pipeline, model and validation
Akram Habib	Faculty co-lead: port use case, review and approvals
Meera Alqubaisi	Reproducibility testing
Mansour Alketbi	Presentation, PDF export and demo video
Mayed Abdulla	Submission checklist and hyperspectral next step

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